Sai Ganesh Chamarty

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Education

University of California, Davis — *Sept 2023 – June 2027 (Expected)*
B.S. in Computer Science (Senior) — *GPA: 3.48 / 4.0*

- **Relevant Coursework:** Data Structures & Algorithms (C++ & Python), Machine Learning & Computer Vision (Python), Computer Architecture, Operating Systems (C/C++), Probability & Statistics, Physics, Multivariable Calculus

Work Experience

Robotics Researcher — *UC Davis LARA Robotics Lab (Davis, CA)* — *Sept 2026 – Present (Incoming)*

- Incoming researcher focused on bimanual robot manipulation using **NVIDIA Isaac Sim**, building on prior self-directed experience with dual-arm robotics and simulation-based validation.

Firmware Developer Intern, AI Chip Performance — *IBM (Poughkeepsie, NY)* — *June 2026 – Sept 2026*

- Developed an **agentic AI** validation framework in Python using **LangGraph** that automatically flags invalid performance runs on **AI accelerators** (Telum, Spyre) in IBM Z mainframe systems, replacing manual review with a consistent, repeatable pipeline.
- Designed and independently built a new agentic architecture that was adopted as the project standard, cutting validation runtime from 15 minutes to 1.5 minutes while doubling detection accuracy; presented results to senior leadership.
- Benchmarked 6 production **AI and ML workloads** (fraud detection, LLM inference with vLLM, AIU utilization) in an **R&D performance initiative**, analyzing tokens/sec and inter-token latency with hardware and firmware teams across multiple countries.

Robotics Systems Engineer & Embedded Infrastructure Lead — *Milpitas, CA and Remote*
Junebase Robotics (Self-Directed) — *Mar 2025 – Present*

- Architected a self-hosted Ubuntu server running **Dockerized Ollama** for local LLM inference and **NVIDIA Isaac Sim**, serving as the compute and storage backbone for server infrastructure, agentic AI, and robotics development.
- Built **agentic AI** workflows (web search, tool-calling, automated documentation) on the self-hosted LLM stack to speed up robot development and cut manual engineering overhead.
- Automated code deployment to NVIDIA Jetson edge devices using **GitHub Actions CI/CD** and Bash pipelines syncing logs and datasets to an 8TB NAS, eliminating manual transfers and enabling rapid hardware-in-the-loop testing.

Projects

Countertop Collaborative Robot — *Junebase Robotics* — Website

- Built a bimanual manipulation platform from two 3D-printed **SO-101** arms, engineering the electrical system (**NVIDIA Jetson Orin Nano**, power delivery, wiring) and low-level Python control and perception software for autonomous pick-and-place.
- Created a digital twin in **NVIDIA Isaac Lab** to validate kinematics and motion sequences before deployment, using Sim2Real alignment to de-risk hardware integration and reduce iteration cycles.
- Engineered a real + synthetic data collection pipeline with automatic formatting and syncing to NAS storage, supporting training of a **Vision-Language-Action (VLA)** model.

Hybrid LLM Inference Architecture — *Junebase*

- Benchmarked **LLM inference** (tokens/sec, inter-token latency, KV cache scaling with context length) across **vLLM**, Ollama, and llama.cpp on cloud GPUs (**NVIDIA A100, L4**) and a local RTX 5070 Ti to characterize price/performance for agentic workloads on the Junebase server.
- Designed a hybrid serving architecture from the results: **Qwen3 14B** locally at 70 tokens/sec for routine agent tasks, offloading high-reasoning tasks to **Gemma4 26B** on an A100 (115 vs. 65 tokens/sec on L4) after cost-per-token analysis.

Skills

- **Languages:** C/C++ (Systems focused), Python, Bash, Swift, CUDA (Exposure)
- **Robotics & Simulation:** NVIDIA Isaac Sim/Lab, Sim2Real Pipelines, Fusion 360, ROS/ROS2 (Learning)
- **AI & Agentic Systems:** Agentic AI (LangGraph), PyTorch, Computer Vision, VLA Models, GR00T N1.5, CNNs, Stereo Vision
- **Systems & Hardware:** AI Chip Performance Engineering, Linux/Ubuntu, NVIDIA Jetson, CI/CD (GitHub Actions), Docker, Ollama, Git, SSH
- **Soft Skills:** Proactive Technical Learning, Cross-functional Collaboration, Technical Documentation, Problem Solving

Leadership & Technical Communities

Founder — Junebase Robotics (Research & Project Lab) — *July 2025 – Present*
Founding President — Swift Coding Club at UC Davis — *Jan 2025 – Present*
Software Director — SacHacks at UC Davis — *June 2024 – Apr 2025*